PAPER_1140: SCm Vacuum Manifold — Mizuno LENR Transmutation Mechanism

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Star-Magic UQFF Framework

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Abstract

Mizuno LENR experiments (Pd–D and Ni–H gas-loading) report anomalous excess heat and elemental transmutation products without the hard radiation expected from nuclear reactions. We derive both phenomena from the SCm Vacuum Manifold using the $F_{U,Bi,i}$ buoyancy field, 1.25 THz phonon resonance, and the 26D Ramanujan amplification factor $S_{26}^{(3)} = 1.4531 \times 10^{26}$.

1 Canonical Constants

Constant	Value	Description
E_{phonon}	8.28×10^{-22} J	THz phonon energy
$S_{26}^{(3)}$	1.4531×10^{26}	26D Ramanujan amplification
Φ_{res}	0.84	Resonance coupling
β_i	0.6	Buoyancy coefficient
κ	5×10^{-4} day^{-1}	SCm decay rate
ρ_{SCm}	7.09×10^{-37} J/m^3	SCm vacuum density
ρ_{UA}	7.09×10^{-36} J/m^3	UA vacuum density

2 Mizuno Excess Heat (SCm)

Mizuno observed 10–300 W excess heat in gas-loaded Ni–D/H systems with anomalous transmutation products (Cu, Cr, Fe from Ni) [1, 2]. The SCm prediction follows the same phonon pathway as Parkhomov but with a lower cluster density N_M:

$$P_{\text{Mizuno}} = N_M \cdot \varepsilon_{\text{cluster}} \cdot e^{-\kappa t} \cdot f_b \tag{1}$$

where $\varepsilon_{\text{cluster}} = 630$ eV $= 1.009 \times 10^{-16}$ J is the canonical Holmlid-calibrated cluster energy [4, 5], and f_b is the system-specific buoyancy stabilisation factor. For $N_M \sim 10^{20}$–10^{21}:

$$\boxed{P_{\text{Mizuno}} \approx 10\text{–}300\ \text{W}} \tag{2}$$

3 Transmutation Mechanism

Standard electroweak theory cannot explain transmutation at low temperatures without MeV-scale particle exchange [3]. SCm provides the mechanism:

1. The 1.25 THz phonon ($E_{\text{phonon}} = 8.28 \times 10^{-22}$ J) drives coherent oscillation of the SCm vacuum manifold within the metal lattice.

2. $F_{U,Bi,i}$ buoyancy modifies the effective nuclear potential barrier via the $\cos(\pi t_n)$ negative-time term, allowing sub-barrier transmutation.

3. The 26D Ramanujan factor $S_{26}^{(3)}$ amplifies the transition probability without requiring high-energy intermediaries.

4. Energy is released into the phonon bath (heat) rather than into particle channels (no hard radiation), consistent with Mizuno's calorimetry [1].

4 Unified LENR Picture

Experiment	System	Observed P	SCm Prediction
Holmlid [4]	K–Fe catalyst	630 eV KER	$\varepsilon_{\text{cluster}} = 630$ eV
Parkhomov [6]	Ni–H, 1100°C	150–280 W	$N \sim 2 \times 10^{18}$, f_b=1
Pons–Fleischmann [7]	Pd–D	1–50 W	V=10^{-6} m^3, f_b=0.001
Mizuno [1]	Ni–D gas	10–300 W	N∼10^{20}–10^{21}, f_b scaled

All four experiments are unified under a single equation (Eq. (1)):

$$P_{\text{LENR}} = N_{\text{eff}} \cdot \varepsilon_{\text{cluster}} \cdot e^{-\kappa t} \cdot f_b(\text{system}) \tag{3}$$

5 Conclusion

SCm provides a single first-principles mechanism — phonon resonance amplified by the 26D vacuum manifold and stabilised by $F_{U,Bi,i}$ buoyancy — that quantitatively reproduces all four major LENR experimental results without ad hoc parameters beyond those already calibrated (κ, $[SSq]$, β_i, Φ_{res}). The unified heat equation (1) with canonical $\varepsilon_{\text{cluster}} = 630$ eV (Holmlid-calibrated [4, 5]) and system-appropriate N_M and f_b spans the full experimental range from Pons–Fleischmann [7] (1–50 W) to Mizuno [1] (10–300 W) and Parkhomov [6] (150–280 W).

References

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